Fiscal Year:	FY 2012	**Task Last Updated:**	FY 10/15/2011
PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	A Novel Biological Countermeasure and Mitigator of High LET-induced Cancer Progression		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
Start Date:	01/01/2011	**End Date:**	12/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Ugur Eskiocak was a doctoral student who has completed his PhD, Sept 2011. He just left the lab to begin a postdoctoral fellowship. Other students are involved in specific aspects of this project (Peter Ly and Lu Zhang). Together with Aadil Kaisani (another graduate student), we have a solid team of students and postdoctoral fellows assisting in this project. Dr. Roig has become an independent assistant professor investigator and is currently working on liver cell cultures and is not longer working as a co-investigator on this project.		
COI Name (Institution):			
Grant/Contract No.:	NNX11AC15G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal addresses Radiation Carcinogenesis Cancer Specific Gap 8: Biological Countermeasures (BCM). While progress has been made since NCRP report 132 in 2000, we still do not have the data for assessing the need for BCMs. However, particle emission levels for fatal cancer risks are projected to be violated for a Mars mission under all possible scenarios. With current permissive exposure limits most astronauts would be limited to about 6-8 months on deep space missions. Thus, if a goal of NASA is to have experienced crews fly multiple times, it is prudent to assume that even with new research to modify uncertainty reductions in fatal cancer risk projection models, that the development of BCMs should now be initiated. The work in progress has been to investigate, Bardoxylone methyl (BARD), an oral available non toxic anti-oxidant inflammatory modulator that is currently in Phase IIB clinical trials for patients with renal insufficiency. BARD is a synthetic triterpenoid and there is a significant molecular understanding of its mode of action. We have already completed investigations with low and high LET irradiation in both normal human lung and colonic epithelial cells and have shown that 10-50nM BARD is an effective radiation BCM, reducing double-strand DNA damage and anchorage-independent cancer progression. Based on our results in human epithelial cells, we now propose to demonstrate BARD is effective in mouse models of cancer. We have a commitment from Reata Pharmaceuticals (Irving, TX) to provide the compound required for animal testing. We believe BARD has an excellent chance of being developed into a potent BCM to protect against high LET irradiation. There are data showing that BARD can protect against radiation-induced mucositis in a hamster cheek pouch assay. In addition, BARD protects against radiation-induced intestinal damage in zebrafish development. Lastly, we have evidence that BARD can mitigate gamma-irradiation (using the colony formation assay) if provided within 30 minutes after IR. In summary, the main objective of this V-NSCOR application is to investigate if BARD is a potent BCM and/or mitigator in mouse models that are predisposed to cancer. Specific aim 1: To use a lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high LET is provided in fractionated/protracted doses over 5 to 10 days (e.g 0.2 Gy x 5 days or 0.1 Gy x 10 days 56Fe 1GeV, compared to single acute 1 Gy 56Fe). Our objective is to determine if pretreatment with BARD prior to acute or fractionated IR reduces the number of invasive (e.g. more lethal) cancers. We will examine at least 3 different ions. Specific aim 2: To test BARD as a BCM in mouse models exposed to SPE simulations. As part of the UT Southwestern Lung NSCOR and the Georgetown GI NSCOR, we have lung cancer and colon cancer susceptible mice that have been treated with solar particle simulations. We propose to test if providing BARD to these animals before irradiation reduces the incidence or progression of cancer. Specific aim 3: To test BARD as a radiation mitigator by providing it to mice within 60 minutes after irradiation. We will compare SPE simulations (total body exposure 2 Gy Protons) and at least two other ions (56Fe at 600 meV, LET 175 and 28Si at 300 MeV LET 70). The significance of the proposed work is that we will be able to demonstrate if BARD reduces cancer incidence or progression in two important tissues (lung and colon) that are susceptible to radiation-associated cancer using doses and doses rates recommended by NASA. This research should provide new data to help reduce uncertainties in current risk projection models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project may lead to the development of a novel radiation countermeasure that will not only be important to protect astronauts from the effects of space radiation, but may also have utility in protecting normal human tissues when radiotherapy is used to treat cancer patients.
Task Progress:	Our project seeks to investigate bardoxolone methyl (BARD), an oral available non toxic anti-oxidant anti-inflammatory modulator that is currently in Phase III clinical trials for type II diabetic patients with renal insufficiency (http://www.reatapharma.com). BARD is a synthetic triterpenoid and there is a significant molecular understanding of its mode of action. We have already completed a series of investigations with terrestrial and space radiation in both normal human lung and colonic epithelial cells and have shown that 10-50nM BARD is an effective radiation biological countermeasure, reducing double-strand DNA damage and anchorage-independent cancer progression. Based on these encouraging results in human epithelial cells, we have now initiated experiments to demonstrate if BARD (CDDO-Me) or it analogue (CDDO-EA) is an effective countermeasure against high-LET irradiation in mouse lung and colon cancer susceptible models. We believe BARD has an excellent chance of being developed into a potent biological countermeasure to protect astronauts against high-LET irradiation. All criteria for approval by the FDA for BARD as a biological countermeasure can be met based on current data except for protection from high-LET irradiation in an animal model. However, there are data showing that BARD does protect against gamma irradiation-induced mucositis in a hamster cheek pouch assay. In addition, BARD protects against gamma irradiation-induced proctitis in a nude rat study. Finally, BARD protects against radiation-induced intestinal damage in zebrafish development. We also have experimental evidence that BARD can mitigate steps in cancer progression from gamma-irradiation (using colony formation assays) if provided 30 minutes after irradiation. In summary, the main objective of this project is to investigate if BARD is a potent biological countermeasure and/or mitigator of SPE simulations and NASA priority HZE ions in mouse models that are predisposed to solid cancers. Since BARD is an anti-inflammatory modulator of oxidative damage, we believe BARD may also protect against HZE induced short-term and long-term neurodegenerative cognitive deficits and perhaps cardiovascular injury. We have already initiated research at the NSRL that support approaches to moving BARD and it analogues specifically, forward as lead space radiation biological countermeasure candidates. Our hypothesis is that BARD may lower the HZE carcinogenic risks. Specific Aim 1: To use the LA-1 lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high-LET is provided in fractionated/protracted doses. Our objective is to determine if pretreatment with non toxic doses of CDDO-EA prior to acute or fractionated HZE irradiation reduces the number of invasive (e.g. more lethal) lung cancers. We will compare this to other priority HZE ions using fluences and energies that are representative of GCR including Silicon and Oxygen. Specific aim 2: To test CDDO-EA as a biological countermeasure in mouse models exposed to SPE (solar particle event) simulations.

	Specific aim 3: To test CDDO-EA as a radiation mitigator by providing it to mice within 30-60 minutes after irradiation (via IP and orally administered gavage). We will compare SPE simulations (total body exposure 2 Gy protons) and two other ions (Iron at 600 MeV, LET 175 at dose ranges between 0.1 Gy and 0.5 Gy and Silicon at 300 MeV LET 70 at doses between 0.2 and 0.8 Gy). We will test single acute versus fractionated doses of both iron and silicon. During the first year of this project we have conducted two and are about to conduct our third series of experiments at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory. In the lung cancer susceptible mouse model we have irradiated approximately 200 mice with single and fractionated doses of iron and 37 of these have been pretreated with the countermeasure CDDO-EA by providing the mice chow with the active ingredient. The incidence and progression of invasive (more lethal) lung cancers will be determined within 6 months. In addition we have exposed 50 mice to a solar particle simulation. Half of these mice were provided the countermeasure in their lab chow prior to the solar particle simulation. These experiments are also ongoing and results should be determined during the Spring 2012. In the colon cancer susceptible mouse model, we have exposed mice to silicon, iron irradiations (single and fractionated doses) and a proton solar particle event simulation. We currently have 80 mice that have been exposed to 2 Gy (of varying energies) of protons and an additional 40 mice have been fed CDDO-EA in their lab chow prior to proton exposure. These are ongoing experiments and we should have initial results within 6 months. In summary, the focus of our proposed research is to provide the basis for improved estimation and uncertainty reduction for solid cancers (lung and colon) from space IR and to develop supporting evidence that biological countermeasures such as CDDO-EA reduce the risk for solid cancer from HZE IR.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Kim SB, Eskiocak U, Cornelius C, Ly P. Biglione S, Shay JW. "Bardoxolone Methyl: A Novel Radiation Countermeasure and Mitigator." Oral presentation (Bardoxylone methyl (CDDO-Me): An antioxidant, anti-inflammatory modulator is a novel radiation countermeasure and mitigator) by J.W. Shay at the IAA Humans in Space Symposium, Houston, Texas, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Kim SB, Eskiocak U, Ly P Kaisani A, Biglione S, Shay JW. "Bardoxylone methyl (CDDO-Me): An antioxidant, anti-inflammatory modulator is a novel radiation countermeasure and mitigator." Poster and oral presentation. 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7030.pdf , Sep-2011
Abstracts for Journals and Proceedings	Kaisani A, Kim S, Delgado O, Batten K, Richardson JA, Wright WE, Story MD, Xie XJ, Minna JD, Shay JW. "Risk Assessment of Radiation Induced Invasive Cancer in Mouse Model of Lung and Colon Cancer." Poster and oral presentation at the 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7120.pdf , Sep-2011
Awards	Kaisani A. "2nd place in the predoctoral poster/oral session for: Kaisani A, Kim S, Delgado O, Batten K, Richardson JA, Wright WE, Story MD, Xie XJ, Minna JD, Shay JW. Risk Assessment of Radiation Induced Invasive Cancer in Mouse Model of Lung and Colon Cancer, 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011." Sep-2011
Awards	Kim SB. "2nd place in the postdoctoral poster/oral session for: Kim SB, Eskiocak U, Ly P Kaisani A, Biglione S, Shay JW, Bardoxylone methyl (CDDO-Me): An antioxidant, anti-inflammatory modulator is a novel radiation countermeasure and mitigator, 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011." Sep-2011